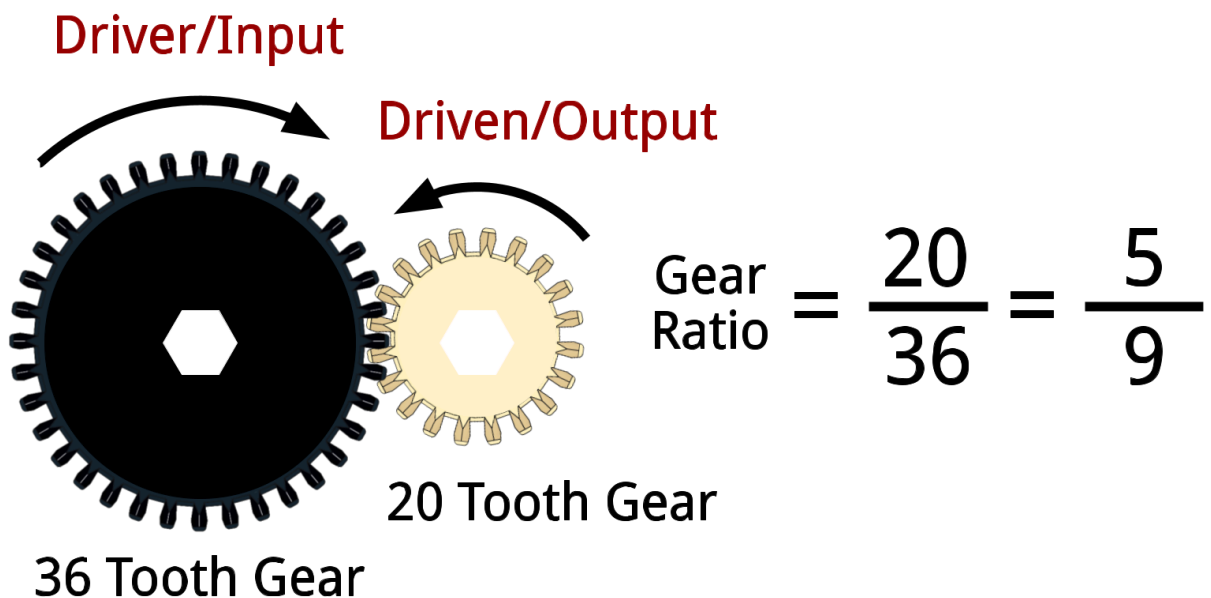


Simple Gear Trains

Gears are mechanisms that are useful in transferring and changing force and motion when something rotates. That transfer is controlled by the size and number of gears and or pulleys. The size and number of them creates something called a **gear ratio**, which is what determines the change in rational force and rotational motion. For mechanisms that rotate we will also use a special name for a force moving in a circle. It is called **torque**. For how fast an object rotates we will use a term called **angular speed**.

To figure out a gear ratio with gears you count the number of gear teeth on an input and then an output gear and create a fraction representing a ratio between those numbers. For example, if I put the two gears below in a **gear train**, which is two or more gears connected together, they have a gear ratio that is a fraction of 20/36 or reduced to 5/9 or 0.55



An input gear is also called the **Driver Gear** and the output gear is also called the **Driven Gear**.

The first and last gears in a gear train will determine the total gear ratio. Any other gears in the chain only serve to change the output direction and are called **Idler Gears**.

The relationship between gear ratios, torque, and angular speed is very straight forward. The gear ratio determines whether driven gear either reduces angular speed and increases torque or reduces torque and increases angular speed. The table below outlines that relationship.

Gear ratios change torque and angular velocity in the following ways.

Gear Ratio	Less than 1	Equal to 1	Greater than 1
Torque	Gets smaller	No change	Gets bigger
Angular Speed	Gets bigger	No change	Gets smaller

The equations for determining angular speed and torques changes from gear ratios are

$$\frac{N_{out}}{N_{in}} = \frac{\tau_{out}}{\tau_{in}} = \frac{\omega_{in}}{\omega_{out}}$$

Where

N_{out} = Number of teeth on the output gear

N_{in} = Number of teeth on the input gear

τ_{out} = Output torque (measured in Newton-Meters M-m or pound-feet or lb-ft)

τ_{in} = Input torque (measured in Newton-Meters M-m or pound-feet or lb-ft)

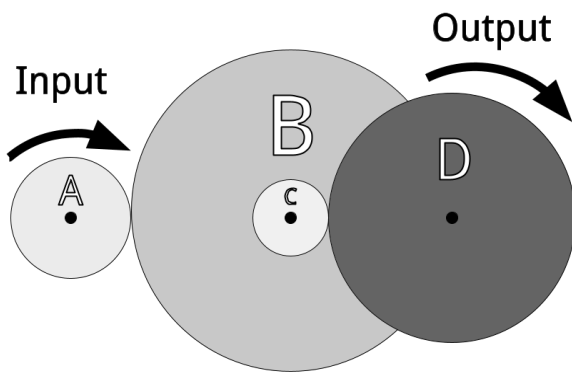
ω_{out} = Output angular velocity (measured in revolutions per minute or rpms)

ω_{in} = Input angular velocity (measured in revolutions per minute or rpms)

Compound Gear Chains

Compound gears are gear trains where one or more axles share multiple gears, allowing for larger angular speed and torque relationships in a smaller mechanism. Gear ratios are calculated by taking the product (multiplying) the gear ratios of the separate connected gear trains. As seen below there are 2 separate gear chains (AB and CD) that are connected by driving the input of the second (CD) on the same axle as the output of the first (AB).

This allows for a substantial change in gearing without using cumbersome large gears. To get the same gear ratio with only 2 gears you would need a 180 tooth gear to be driven by the 18 tooth gear. It would be 3 times the circumference of the largest gear used below which has 60 teeth.



Gear B and C share an axle.

A=18 Teeth C=12 Teeth

B=60 Teeth D=36 Teeth

$$\text{Gear Ratio} = \frac{B}{A} \times \frac{D}{C} = \frac{B \times D}{A \times C}$$

$$\frac{60}{18} \times \frac{36}{12} = \frac{60 \times 36}{18 \times 12} = \frac{2160}{216} = \frac{10}{1}$$

$$\text{Gear Ratio} = 10$$